Experiment 19

Discrete-Time LTI System Realization, FIR/IIR Filtering and Frequency Response

OBJECTIVES

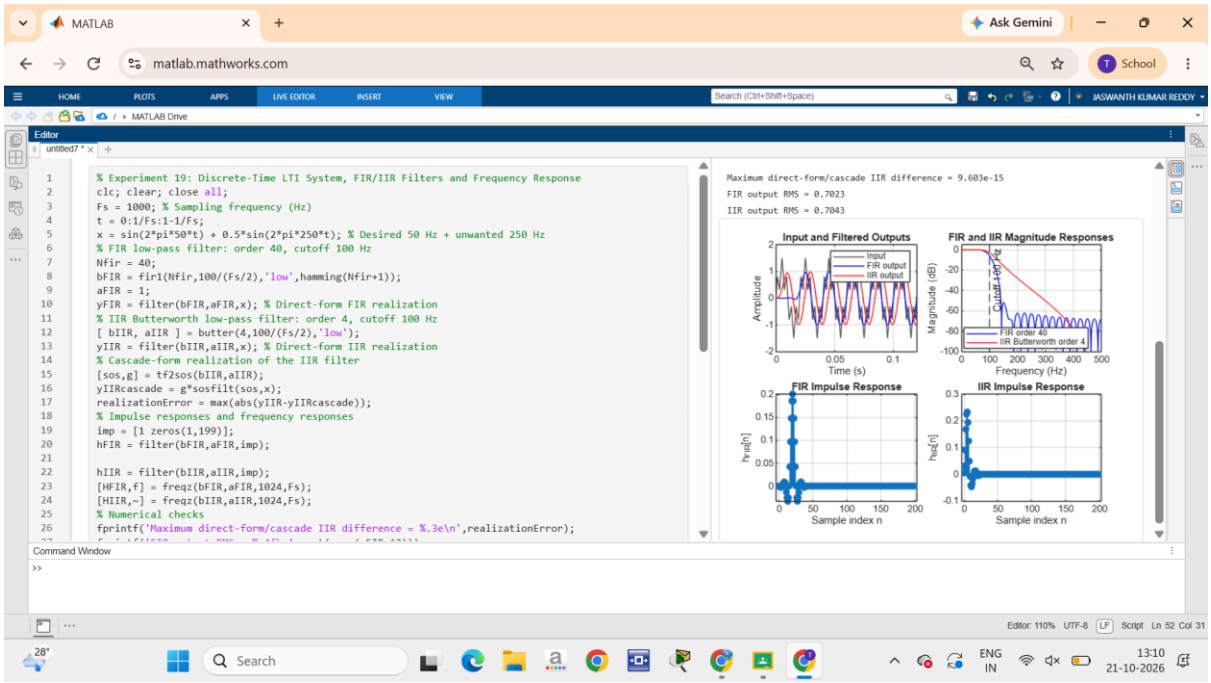
• To analyze a discrete-time LTI system using its transfer function.

• To design and implement FIR and IIR filters for a specified cutoff frequency.

• To compare direct-form and cascade-form realizations of an IIR system.

• To determine impulse and frequency responses and evaluate filtering performance from input/output signals.

Output Graphs:



Result: Discrete-time LTI systems were implemented using realization structures, and FIR and IIR filters were analyzed through their input-output responses and frequency characteristics.